



ELIAS FUNDI NJERU

Geomatics Engineer | GIS Specialist | Backend Developer

Siakago, Embu County, Kenya | +254 792 201 418 | eliasfundi6@gmail.com

PROFESSIONAL SUMMARY

Geomatics Engineer with a B.Sc. from Dedan Kimathi University of Technology (Second Class Honours, Upper Division, 2025). I work across cadastral surveying, GIS analysis, and backend development. I pick up new tools quickly, enjoy solving spatial problems with software, and am looking to grow with a team that values practical skills and initiative.

TECHNICAL SKILLS

Surveying Equipment: Total Station, GNSS/RTK, Theodolite, Dumpy Level, GPS Receivers, EDM

GIS & Mapping: ArcGIS Pro, QGIS, Google Earth Pro, AutoCAD Civil 3D, ArcGIS Online

Survey Software: Trimble Business Center, Leica Geo Office, Topcon Magnet

Programming: Python (GeoPandas, Rasterio, Shapely, PyProj), C#, .NET / ASP.NET Core

Backend & Web Dev: REST API development, ASP.NET Core, GitHub Pages, web application deployment

DevOps & Cloud: Kubernetes, Docker (fundamentals), Cloud Computing concepts

Remote Sensing: Satellite Image Analysis, Drone Mapping, Photogrammetry, DEM Generation

Data & Computations: Triangulation & Traverse Adjustments, COGO, Least Squares Adjustment

Productivity: Microsoft Office Suite, Database Management, Technical Report Writing

PROFESSIONAL EXPERIENCE

Survey Technician | *Mbeere North Sub-County Survey Office, Siakago*

May 2023 – Present

- Assist licensed surveyors in cadastral, boundary, and topographical surveys across Mbeere North
- Operate Total Station and GNSS/RTK equipment for control network establishment and traversing
- Perform survey computations: traverse adjustments, COGO, UTM Zone 37S coordinate transformations
- Produce official Ministry of Lands boundary determination reports and surveyor correspondence
- Built a local offline Streamlit automation app ("Ruth") using Ollama/LLaMA 3.2 to reduce repetitive letter drafting for the sub-county office
- Manage a private client tracking system covering parcel numbers, fee schedules, and payment milestones

Planning AttachÃ© – Surveying & GIS | *Murang'a County Government, Dept. of Physical Planning*

Jan 2024 – Apr 2024

- Led field surveys and GPS data collection for county-wide market plots digitization exercise
- Served as data analyst for the assets valuation project, managing spatial databases and land records
- Prepared Part Development Plans and Local Physical and Land Use Development Plans
- Conducted resource inventorying and topographical assessments for infrastructure planning
- Performed QC on cadastral and boundary survey data ensuring compliance with national standards

Freelance GIS & Surveying Tutor | *Remote – Students in Kenya, US & Canada*

Mar 2023 – Present

- Deliver practical training in QGIS, ArcGIS Pro, AutoCAD Civil 3D, and spatial analysis workflows

- Mentor 10+ students through GIS assignments, crime analysis, i2 Analyst's Notebook, and SPSS
- Develop tutorials on coordinate transformations, map production, and road alignment design
- Support students at institutions including Seneca College (Canada) with GIS coursework delivery

Tech Trainer | *Campus Digital Kenya (EdTech Initiative)*

2024 – Present

- Train youth in digital and technology skills as part of the Campus Digital Kenya EdTech programme
- Contribute to grant proposals and curriculum development for tech skill-building in East Africa

RESEARCH & ACADEMIC PROJECTS

HydroGPT – AI-Powered Water Accessibility Platform |

2024 – 2025

BSc Final Year Research Project, DeKUT

- Designed and built a full-stack web platform analysing water point accessibility for 130,000+ residents of Mbeere South Sub-County, Kenya
- Implemented network-constrained spatial modelling using PostGIS and pgRouting to compute service coverage across 147 water points
- Integrated Claude AI as a natural-language query interface, enabling non-GIS users to interrogate spatial data conversationally (96% query accuracy)
- Built with FastAPI backend, PostgreSQL/PostGIS, React.js + Leaflet frontend; submitted as BSc dissertation (Second Class Honours, Upper Division)

EDUCATION

B.Sc. Geomatics Engineering & Geospatial Information Systems

Second Class Honours (Upper Division)

Dedan Kimathi University of Technology (DeKUT), Nyeri

Graduated: December 2025

Minor: Critical Thinking and Communication Skills

Kenya Certificate of Secondary Education (KCSE) – Mean Grade B Plain | *Siakago Mixed Day Secondary School, 2020*

CERTIFICATIONS & TRAINING

- **Landslide Monitoring & Risk Assessment Using NASA Earth System Data** – *NASA ARSET Programme* (March 2025)
- **Calculating Disparities in Pollution Exposure Using ArcGIS** – *Esri Training (ArcGIS Lab)* (January 2024)
- **C# Full Course – Multi-threading, OOP, Backend Concepts** – *Bro Code (YouTube)* (2025)
- **Back-end Web Development with .NET for Beginners** – *Microsoft dotnet (YouTube/Learn)* (2025)
- **Kubernetes Crash Course for Absolute Beginners** – *TechWorld with Nana (YouTube)* (2025)

KEY COMPETENCIES

- Cadastral & Geodetic Surveying
- Topographical Survey & Mapping
- GIS Analysis & Spatial Data Management
- Survey Report Writing & Documentation
- Land Acquisition & Boundary Dispute Resolution
- C# / .NET Backend Development
- Python Scripting & Automation
- Kubernetes Fundamentals & Container Basics
- Remote Sensing & Satellite Image Analysis
- Team Leadership & Client Management

PROFESSIONAL AFFILIATIONS

- Institution of Surveyors of Kenya (ISK) – Student Member
- Esri Training Community – Certified Spatial Analysis Practitioner
- NASA ARSET Programme Alumnus

PROFESSIONAL REFEREES

Available on request.